type '1.' Repeat the above procedure with But_OFF, and in the text box, type '0.'

Finally, click on But_Close and select When But_Close.Click. Now, click on an empty space in the viewer pane and start typing Close. A list will open, and from the list, select Close application. Match it in the empty socket.

This completes the design setting of the device. The complete picture is shown in Fig. 10.

It is now time to install the program on the mobile. Go to Play Store, download, and install MIT AI2 companion (if it has not been done already). Now go to the computer screen, click on Build (at the top of the screen), and select Android App (.apk). The program will compile, and a QR code will appear. On the mobile, open MIT AI2 Companion and scan the QR code. Follow the instructions on the screen (ignore warnings – there is no malicious code here) and install the app. (Ensure both the mobile and computer are on the same Wi-Fi network.) Now, turn on the HC-05 Arduino combination and press Scan on the mobile. A new screen will appear, showing the name 'HC-05' and its MAC address. Click on it, and the screen will return to the original, allowing 'Connect' to be pressed.

Testing

Power the Arduino using USB. If the screen that appears when Scan is pressed is blank, check and adjust the app permissions. Go to Settings and ensure Bluetooth is enabled. Also, go to Settings > Apps > Permissions > Other Permissions > Apps and grant all necessary permissions to the app.

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

(Some mobiles may have slightly different names, but the procedure is similar.) This completes the procedure for Bluetooth control through an Android mobile. EFY

Hemakumar K., with 27 years of teaching and other experience, retired as an Associate Professor from the Government College of Engineering, Kannur, Kerala.